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Course –ECA5601- 5g/6g Communication

Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

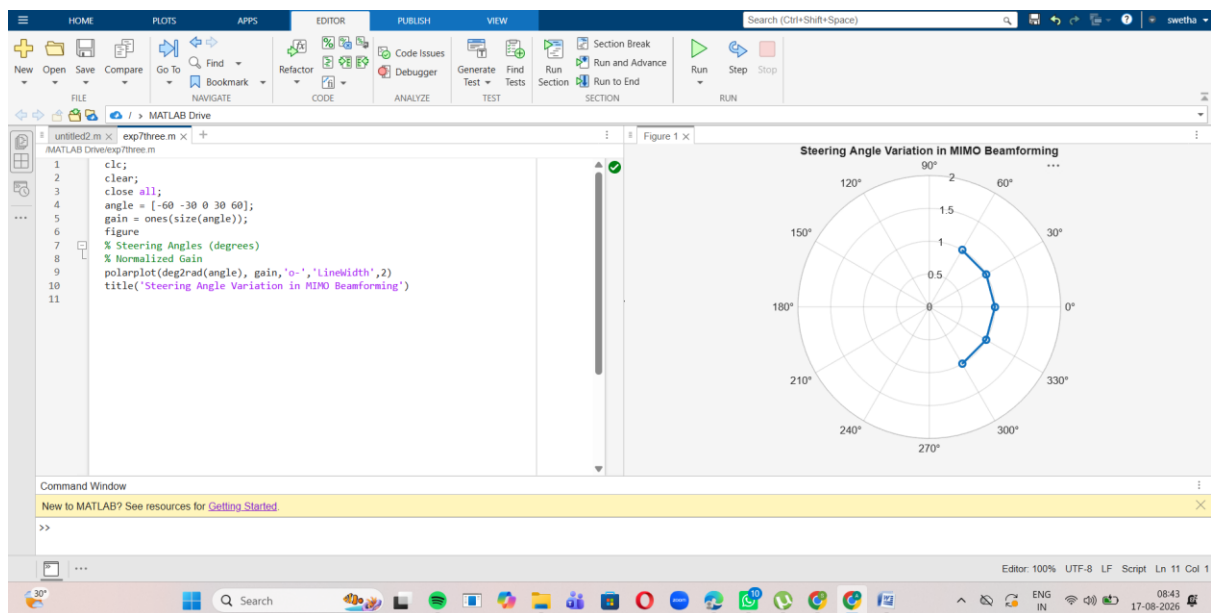
Procedure

1. Define the MIMO system configuration (e.g., number of transmit and receive antennas such as 2×2 or 4×4).
2. Generate random binary data bits to represent the transmitted signal.
3. Modulate the bits using a digital modulation scheme (e.g., BPSK or QPSK).
4. Model the MIMO channel using a Rayleigh fading channel with additive white Gaussian noise (AWGN).
5. Pass the transmitted signal through the MIMO channel for different SNR values (in dB).
6. At the receiver, apply detection techniques (e.g., Zero-Forcing or Maximum Likelihood detection).
7. Demodulate the received signal to recover transmitted bits.
8. Compute Bit Error Rate (BER) by comparing transmitted and received bits.
9. Calculate Channel Capacity (bps/Hz) using SNR and MIMO capacity formula.
10. Plot BER vs SNR and Capacity vs SNR graphs for performance analysis.

objective 1 :

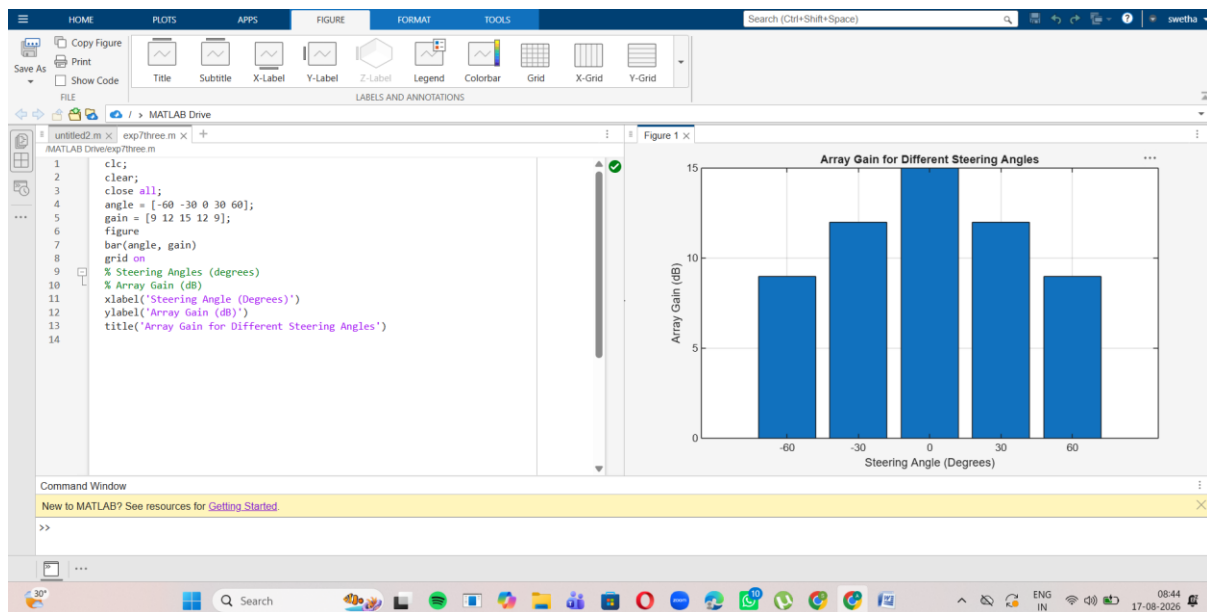
Program

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60];
gain = ones(size(angle));
figure
% Steering Angles (degrees)
% Normalized Gain
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2)
title('Steering Angle Variation in MIMO Beamforming')
output
```



Objective 2: code and output:

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60];
gain = [9 12 15 12 9];
figure
bar(angle, gain)
grid on
% Steering Angles (degrees)
% Array Gain (dB)
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```

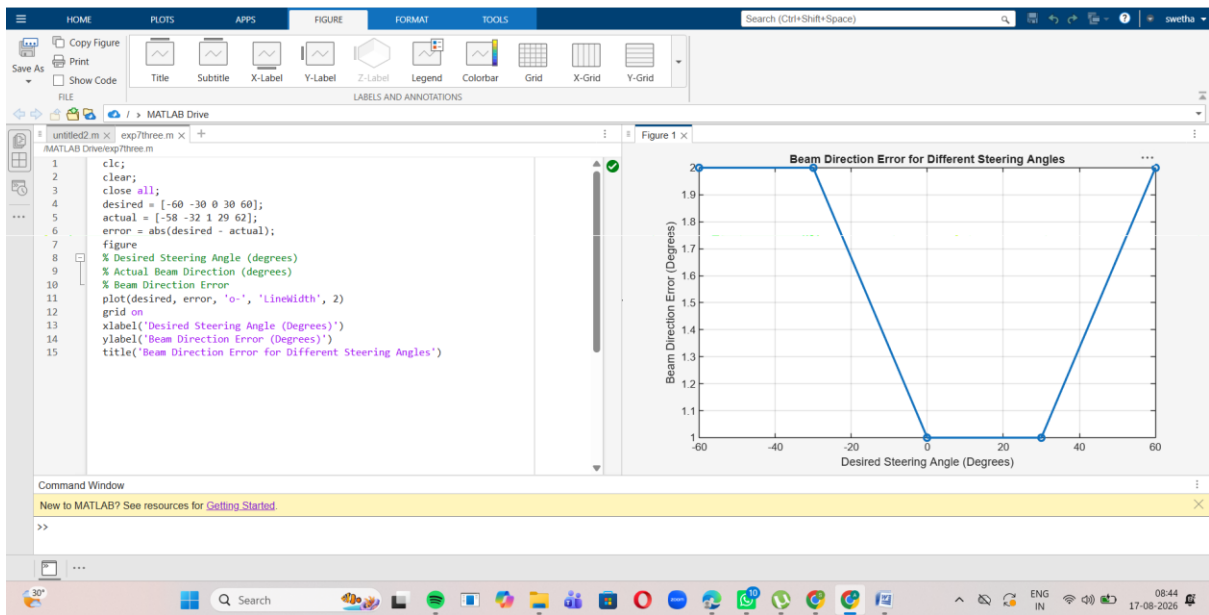


Objective 3:code and output:

```

clc;
clear;
close all;
desired = [-60 -30 0 30 60];
actual = [-58 -32 1 29 62];
error = abs(desired - actual);
figure
% Desired Steering Angle (degrees)
% Actual Beam Direction (degrees)
% Beam Direction Error
plot(desired, error, 'o-', 'LineWidth', 2)
grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')

```



Result:

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.